

TAYYABA ANWAR

Software Engineer | Full Stack Developer

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SUMMARY

Results-driven Software Engineer with a strong foundation in full-stack development and intelligent system integration. Proficient in building scalable web applications using React, Angular, FastAPI, and ASP.NET Core. Expert in database management with PostgreSQL, MySQL, and SQL Server, and experienced in deploying production-ready AI-powered features. Committed to clean code, performance optimization, and staying at the forefront of modern development tools like Cursor, Claude, and Codex.

EDUCATION

Punjab University College of Information Technology (PUCIT)

Bachelor of Science in Information Technology | CGPA: 3.49/4.0

Lahore, Pakistan

Dec 2022 – May 2026

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), C#, Java, SQL, C++, HTML5, CSS3

Frontend: React.js, Angular, Tailwind CSS, Framer Motion, Zustand

Backend: FastAPI, ASP.NET Core, RESTful APIs

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, ChromaDB (Vector Store)

Modern Workflow & AI: Cursor, Claude, Codex, Ollama, LangChain, Git, GitHub, AWS, Docker

Core Competencies: Full-Stack Architecture, System Design, DSA, RAG Pipeline Design

EXPERIENCE

Arch Technologies

Full Stack Developer Intern

Lahore, Pakistan

Dec 2025 – Jan 2026

- Successfully developed and delivered complex full-stack features for multiple projects while strictly adhering to rigorous production deadlines.
- Leveraged end-to-end technical expertise to architect scalable software solutions and ensure seamless cross-platform performance.

PROJECTS

EduVerse AI – AI-Powered Multi-Tenant E-Learning Platform (Capstone FYP) | Angular, FastAPI, MongoDB, LangChain, ChromaDB, AWS

- Engineered end-to-end RAG pipeline using LangChain: document ingestion (PDF/PPTX/DOCX), recursive text chunking, embedding with all-MiniLM-L6-v2, and semantic retrieval via ChromaDB vector store
- Integrated multiple local LLMs via Ollama (phi3.5, qwen2.5:3b, llama3.2:3b) with task-specific model routing — lesson generation, MCQ creation, and conversational AI tutoring
- Built a student performance classifier analysing quiz scores and response times to adaptively generate personalised lesson content at Slow / Average / Fast pace levels
- Implemented a two-layer hallucination detection pipeline using BERTScore (semantic similarity) and ROUGE (lexical overlap) to validate AI output against source documents
- Designed multi-tenant SaaS architecture with secure tenant-based data isolation and independent institutional branding on AWS

View Code: <https://github.com/Muhammad-Hassan-522/EduVerse-AI-backend.git>

SnapSecure AI – Privacy-First Redaction Tool | FastAPI, React, Tesseract OCR, OpenCV

- Developed AI-powered sensitive information detection pipeline using Tesseract OCR and OpenCV to redact SSNs, emails, and phone numbers from images with 95%+ accuracy
- Built real-time image processing backend with FastAPI and React frontend, deployed on cloud infrastructure

View Code: <https://github.com/tayyaba13-02/SnapSecure-AI>

Live Demo: <https://huggingface.co/spaces/tayyabaaly/SnapSecureAI>

TailorNow | Service Marketplace Platform | ASP.NET MVC, C#, SQL Server, JavaScript

- Developed end-to-end tailoring marketplace with real-time booking system and role-based dashboards for customers, tailors, and admins
- Designed normalized SQL Server database schema and implemented Entity Framework ORM for efficient data management

View Code: <https://github.com/tayyaba13-02/TailorNow>

Live Demo: <https://huggingface.co/spaces/tayyabaaly/TailorNow>

ACHIEVEMENTS & CERTIFICATIONS

- TECH WAR Coding Competition: Recognized for algorithmic problem-solving and DSA proficiency.
- Open Source Contributions: Active GitHub profile with 5+ production-grade repositories.
- Academic Excellence: Consistently maintained a high CGPA in a rigorous BSIT curriculum.